Sub Code: 23CS2T04,23IT2T04,23AD2T04,23AM2T04 **MIC23**

B. Tech II Semester Regular Examinations

DATA STRUCTURES

(CSE, AI&DS, AI&ML, IT)

(Model paper-I)

Time: 3 hours **Max Marks: 70M**

Note: 1. Question paper consists of two parts (**Part A and Part B**)
2. Answer all sub questions from **Part A**
3. Answer all questions from **Part B** with **either or choice**

		PART – A			**10 X 2 = 20 M**
			CO	**BTL**	**Marks**
I.	**a)**	Define Data Structure. Give Examples.	CO1	**L2**	**2 M**
	b)	What is the complexity of linear search.	CO1	**L1**	**2 M**
	c)	Explain the advantages of linked list over arrays.	CO2	**L2**	**2 M**
	d)	Write the structure of linked list which stores the list of students.	CO2	**L3**	**2 M**
	e)	Explain Stack overflow and underflow condition.	CO3	**L2**	**2 M**
	f)	What are the application of stack.	CO3	**L1**	**2 M**
	g)	What are the types of queues.	CO4	**L1**	**2 M**
	h)	What are the advantages of circular queue over linear queue implemented using arrays.	CO4	**L2**	**2 M**
	i)	What is an expression tree. Explain with example.	CO5	**L3**	**2 M**
	j)	What is a hash function.	CO5	**L1**	**2 M**
		PART – B			**5 X 10 = 50 M**
		UNIT – I			
1.	**a)**	Explain the classification of data structures.	CO1	**L2**	**5 M**
	b)	Write an algorithm for searching an element in an array using Binary Search . Explain with an example.	CO1	**L1**	**5 M**
		(OR)			
2.	**a)**	Sort the followitn elements using Insertion sort: 12, 3,6,7,4,9. Write the algorithm for Insertion sort.	CO1	**L4**	**5 M**
	b)	Write an algorithm for merge sort. Explain with an example.	CO1	**L2**	**5 M**
		UNIT – II			
3.		What is a linked list? Explain the operations on Single Linked List.	CO2	**L2**	**10 M**
		(OR)			
4.		What is a Circular Linked List? Explain the operations on Circular Linked List.	CO2	**L2**	**10 M**
		UNIT – III			
5.		What is a stack? Explain the linked list implementation of Stack.	CO2	**L2**	**10 M**
		(OR)			
6.	**a)**	Write an algorithm for converting infix expression to postfix expression.	CO3	**L2**	**5 M**
	b)	Convert the following expression into postfix: (A*B)+(C/D)-(D+E)	CO3	**L5**	**5 M**
		UNIT – IV			
7.	**a)**	What is a queue? Explain the operations on queue.	CO4	**L2**	**5 M**
	b)	Explain the implementtation of the queue using Arrays.	CO4	**L2**	**5 M**
		(OR)			
8.		What is a Deque. Explain the implementation of deque using circular arrays.	CO4	**L2**	**10 M**
		UNIT – V			
9.	**a)**	What is a binary tree? Explain the representation of binary	CO5	**L1**	**5 M**

		tree in memory.			
	b)	Explain preorder, postorder and inorder traversal of a binary tree.	CO5	**L3**	**5 M**
(OR)					
10.	**a)**	What is a hash function? Explain Division and Multiplication method.	CO5	**L2**	**5 M**
	b)	Explain Collision resolution by open addressing.	CO5	**L3**	**5 M**

***** End of the Question Paper *****

Sub Code: 23CS2T04,23IT2T04,23AD2T04,23AM2T04 **MIC23**

B. Tech II Semester Regular Examinations

DATA STRUCTURES

(COMMON TO CSE,AIDS,AIML,IT)

(Model Paper-II)

Time: 3 hours **Max Marks: 70M**

Note: 1. Question paper consists of two parts (**Part A and Part B**)
2. Answer all sub questions from **Part A**
3. Answer all questions from **Part B** with **either or choice**

		PART – A			**10 X 2 = 20 M**
			CO	**BTL**	**Marks**
I.	**a)**	What are primitive and non primitive data structures	CO1	**L2**	**2 M**
	b)	What are the advantages of binary search over linear search	CO1	**L1**	**2 M**
	c)	What is a circular linked list	CO2	**L2**	**2 M**
	d)	Write the polynomial representation of $5x^8+3x^5-7x^2+7x-3$	CO2	**L3**	**2 M**
	e)	What is a stack?	CO3	**L2**	**2 M**
	f)	Evaluate the postfix expression 5 3 * 6 + 2 + using stack	CO3	**L1**	**2 M**
	g)	What are the application of queues	CO4	**L1**	**2 M**
	h)	What is a Binary Search Tree.	CO4	**L1**	**2 M**
	i)	What is a complete binary tree	CO5	**L3**	**2 M**
	j)	How is hash table better than direct access table	CO5	**L1**	**2 M**
		PART – B			**5 X 10 = 50 M**
		UNIT – I			
1.	**a)**	Explain Time Complexity and Space Complexity	CO1	**L2**	**5 M**
	b)	Write an algorithm for linear search. Explain with an example	CO1	**L1**	**5 M**
		(OR)			
2.	**a)**	Sort the following elements using selection sort: 5 ,3, 2, 6, 1,4	CO1	**L3**	**5 M**
	b)	Write an algorithm for quick sort. Explain with an example.	CO1	**L2**	**5 M**
		UNIT – II			
3.	**a)**	What is a Single Linked List? Expain the deletion operation on single linked list	CO2	**L2**	**5 M**
	b)	Write the algorithm for inserting an element at the starting and ending of circular linked list.	CO2	**L2**	**5 M**
		(OR)			
4.	**a)**	What is a double linked list? Explaing the operations of Double linked list	CO2	**L3**	**10 M**
		UNIT – III			
5.	**a)**	What is a stack? Explain the operations on stack using arrays	CO3	**L2**	**5 M**
	b)	Write an algorithm for reversing a list.	CO3	**L3**	**5 M**
		(OR)			
6.	**a)**	Write the algorithm for evaluating a postfix expression.	CO3	**L2**	**5 M**
	b)	Evaluate the postfix expression 9 3 4 * 8 + 4 / -	CO3	**L3**	**5 M**
		UNIT – IV			
7.	**a)**	Explain the implementation of queue using linked list	CO4	**L2**	**10 M**
		(OR)			
8.	**a)**	Expain Breadth first Search using Queues	CO4	**L2**	**5 M**
	b)	Explain the operations of deque.	CO4	**L2**	**5 M**
		UNIT – V			
9.	**a)**	What is a binary search tree? Construct binary search tree for 45,25,36,47,34,58,63,2,6,72	CO5	**L5**	**5 M**
	b)	Write an algorithm for deleting a node in BST	CO5	**L2**	**5 M**
		(OR)			
10.	**a)**	What is a hash function? Explain different hash functions	CO5	**L2**	**5 M**
	b)	Explain Collision resolution using chaining	CO5	**L3**	**5 M**